From Classroom/Integrated School Programs 7 in Top 20, 23 in Top 100, 54 in Top 300, 106 in Top 500 All India Ranks & 2314 Students from Classroom /Integrated School Programs & 3723 Students from All Programs have been Awarded a Rank in JEE (Advanced), 2013

ALL INDIA TEST SERIES

FIITJEE

JEE (Advanced), 2014

FULL TEST – V

Paper 2

Time Allotted: 3 Hours **Maximum Marks: 237**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part is further divided into three sections: **Section-A, Section-B & Section-C**
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

(i) **Section-A (01 – 06)** contains 6 multiple choice questions which have only one correct answer. Each question carries **+5 marks** for correct answer and **– 2 marks** for wrong answer.
Section-A (07 – 12) contains 2 paragraphs. Based upon paragraph, 3 multiple choice questions have to be answered. Each question has only one correct answer and carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

(ii) **Section-B (01 – 02)** contains 2 Matrix Match Type questions containing statements given in 2 columns. Statements in the first column have to be matched with statements in the second column. Each question carries **+8 marks** for all correct answer. For each correct row **+2 marks** will be awarded. There may be one or more than one correct choice. No marks will be given for any wrong match in any question. There is no negative marking.

(iii) **Section-C (01 – 05)** contains 5 Numerical based questions with single digit integer as answer, ranging from 0 to 9 and each question carries **+3 marks** for correct answer. There is no negative marking.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

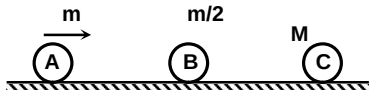
PART – I

SECTION – A

Straight Objective Type

This section contains **6 multiple choice questions** numbered 1 to 6. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.

1. Three masses A, B and C are kept on the smooth horizontal surface as shown in the figure. A sharp impulse is given to the mass A, so that it starts moving towards B with speed v_0. What is the minimum value of m so that there is only one collision between masses A and B. (all collision are elastic)



(A) M

(B) $\frac{10M}{3}$

(C) $\frac{8M}{3}$

(D) 2M

2. A freshly prepared sample contains 16×10^{20} radioactive nuclei, whose mean life is 10^{10} seconds. The activity of the sample just after 4 half lives time is

(A) $\frac{\ln^2}{16} \times 10^{10}$ dps

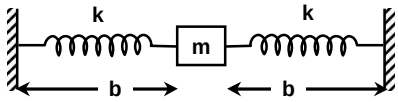
(B) 1×10^{10} dps

(C) $\frac{16}{\ln^2} \times 10^{10}$ dps

(D) 16×10^{10} dps

Space for rough work

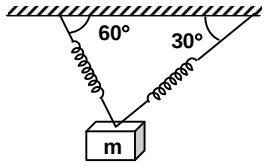
3. In the adjacent figure, a block of mass m is rigidly attached to two identical springs of stiffness k each. The other ends of the springs are connected to the fixed wall when the block is in equilibrium, length of each spring is b, which is greater than then the natural length ℓ of the spring The time period of the oscillation of the block if it is displaced by small distance x perpendicular to the length of the spring and released. (motion of the block is unaffected from gravity).



(A) $2\pi\sqrt{\frac{mb}{k(b-\ell)}}$ (B) $2\pi\sqrt{\frac{mb}{2k(b-\ell)}}$

(C) $2\pi\sqrt{\frac{m(b-\ell)}{kb}}$ (D) $2\pi\sqrt{\frac{m(b-\ell)}{2kb}}$

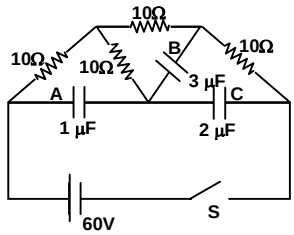
4. Block of mass m = 10 kg is suspended by two springs as shown in the diagram. Acceleration of block just after cutting the right spring is



(A) $\frac{5\sqrt{3}}{2}$ m/s^2 (B) 10 m/s^2

(C) $5\sqrt{3}$ m/s^2 (D) 5 m/s^2

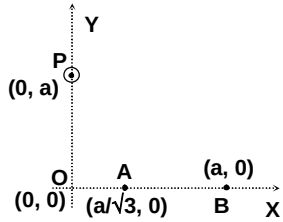
5. In the adjacent figure the switch S is closed at t = 0. Q_A, Q_B and Q_C are charges on the capacitor A, Capacitor B and capacitor C in the steady state respectively. Choose the correct statement.



(A) $Q_A < Q_B < Q_C$ (B) $Q_A < Q_C < Q_B$

(C) $Q_C < Q_A < Q_B$ (D) $Q_C < Q_B < Q_A$

6. An infinite current carrying conductor, parallel to z-axis is situated at point P as shown in the figure to x-y plane. The find $\int_A^B \vec{B}\cdot\vec{d\ell}$?



(A) $\frac{\mu_0 i}{24}$ (B) $\frac{\mu_0 i}{16}$

(C) $\frac{\mu_0 i}{12}$ (D) $\frac{\mu_0 i}{8}$

Space for rough work

Comprehension Type

This section contains **2 groups of questions.** Each group has 3 multiple choice question based on a paragraphs. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** is correct.

Paragraph for Questions 7 and 9

An experiment is performed on thin lens in order to determine the radii R_1 and R_2 of the curvature of its two surfaces and the refractive index 'μ' of its glass.

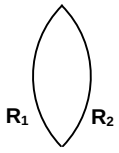
Experiment I: A real object is placed 36 cm to the left of lens and a real image is formed 72 cm to the right of lens.

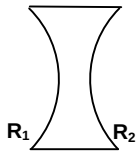
Experiment II: The lens is floated on the surface of mercury with surface having radius of curvature R_2 faces air. A real object is placed at 60 cm above it, and a real image is formed at 30 cm above it (assume mercury to be perfectly reflecting).

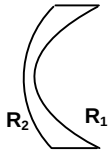
Experiment III: The lens is floated on the surface of mercury with surface having radius of curvature R_1 faces air. A real object is placed at 7.5 cm above it and real image is formed at $\frac{20}{3}$ cm above it. On the bases of Experiments I, II and III, answer the following questions.

7 The refractive index of material of lens is
(A) 1.6 (B) 1.75
(C) 2 (D) none

8 The magnitude of R_1 is
(A) 60 cm (B) 30 cm
(C) 120 cm (D) 45 cm

9 The lens can be best represented by
(A) 

Biconvex
(B) 

Biconcave
(C) 

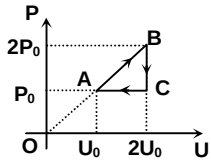
Convexo-concave
(D) 

Concavo-convex

Space for rough work

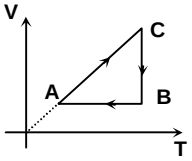
Paragraph for Questions 10 and 12

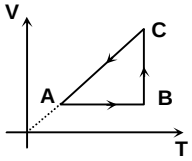
2 mole of an ideal monoatomic gas undergoes a cyclic process ABCA as shown in the figure. R is the universal gas constant. (given ln2 = 0.693)

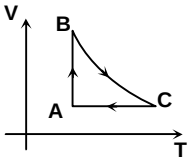


Answer the following questions.

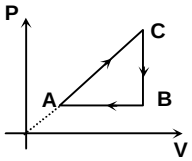
10. Percentage efficiency of cycle is (approximately)
(A) 10.38% (B) 13.38%
(C) 40.38% (D) none of these

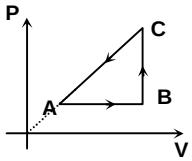
11. Graph of volume verses temperature of the cyclic process ABCA is
(A) 

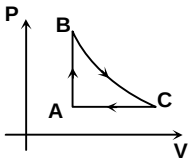
(B) 

(C) 

(D) none of these

12. Graph of pressure verses volume of the cyclic process ABCA is
(A) 

(B) 

(C) 

(D) none of these

Space for rough work

SECTION – B

(Matching List Type)

This section contains **2 multiple choice questions.** Each question has matching Column(s). The codes for the Column(s) have choices (A), (B), (C) and (D) out of which only **ONE OR MORE is/are correct.**

1. An electron in a hydrogen atom makes a transition $n_1 \rightarrow n_2$, where n_1 and n_2 are the principal quantum numbers of the two states. Assume Bohr model to be valid.

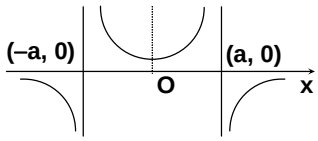
	Column I		Column II
(A)	The electron emits an energy of 2.55 eV	(p)	$n_1 = 2, n_2 = 1$
(B)	Time period of the electron in the initial state is eight times that in the final state.	(q)	$n_1 = 4, n_2 = 2$
(C)	Speed of electron become two times	(r)	$n_1 = 5, n_2 = 3$
(D)	Radius of orbit of electron becomes 4.77A°	(s)	$n_1 = 6, n_2 = 3$
		(t)	$n_1 = 8, n_2 = 4$

Space for rough work

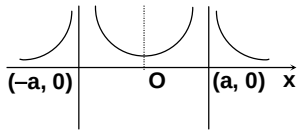
2. The **Column-I** is having the graph of either electrostatic field or its magnitude verses position on x-axis and magnetic field or its magnitude verses position on the x-axis for the system indicated in the **Column-II**. Match the following.

Column I

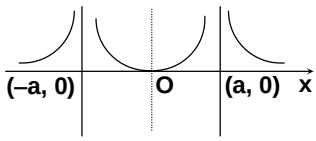
(A)



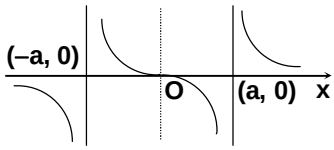
(B)



(C)

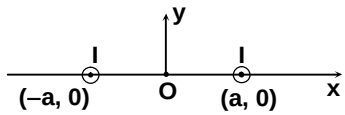


(D)



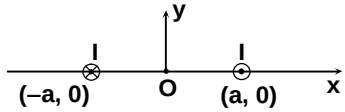
Column II

(p)



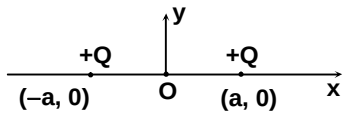
Two infinite wire are kept parallel to z-axis as shown in the figure, each carrying positive current I along positive z-axis

(q)



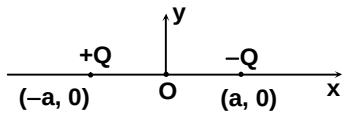
Two infinite wire are kept parallel to z-axis as shown in the figure, one is carrying current positive I along positive z-axis and other is carrying positive current I along negative z-axis

(r)



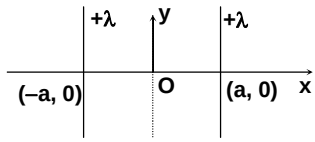
Two point charges, each of magnitude +Q are kept at points (–a, 0) and (a, 0)

(s)



Two point charges, +Q and –Q are kept at points (–a, 0) and (a, 0) respectively

(t)



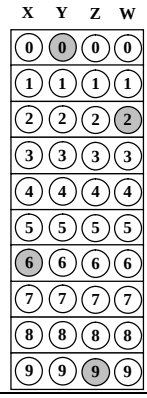
Two infinite non-conducting rod are kept parallel to the y-axis as shown in the figure, each carrying uniform charge density λ.

Space for rough work

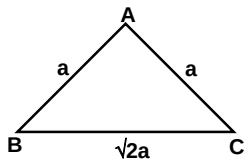
SECTION – C

Integer Answer Type

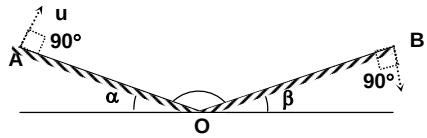
This section contains **5 questions.** The answer to each of the questions is a single-digit integer, ranging from 0 to 9. The appropriate bubbles below the respective question numbers in the ORS have to be darkened. For example, if the correct answers to question numbers X, Y, Z and W (say) are 6, 0, 9 and 2, respectively, then the correct darkening of bubbles will look like the as shown.



1. ABC is an isosceles triangular sheet of negligible thickness, made of isotropic material. Mass of the sheet is m. Its moment of inertia about axes perpendicular to its plane and passing through the points A and B are I_1 and I_2 respectively. [AB = AC = a and BC = $\sqrt{2}$a]. Find I_2/I_1.



2. Two inclined planes OA and OB of inclinations to the horizontal are α and β, each equal to 30° as shown in the figure. A particle is projected at an angle of 90° with plane OA from point A and its strikes the plane OB at point B normally. Then find the speed of projection in m/s. (given that OA = OB = 20 cm and g = 10 m/s^2)



3. If the sound heard by observer, whose equation is given as
y = 8sin10πt cos 200πt at x = 0
The number of beat frequency heard by observer is 2k, then find the value of k.

4. A certain amount of a mono-atomic ideal gas undergoes a process ρu^{η} = C, where ρ is the density of the gas and u is the internal energy of the gas. It was found that the ratio r = $\frac{\Delta W}{\Delta Q}$ for the process was r = 2/3. What is the value of η?

5. Current in a X-ray tube operating at 40 kV is 10 mA. 1% of the total kinetic energy of electrons hitting the target is converted into X-rays. Then the heat produced in the target per second is 99n Joule

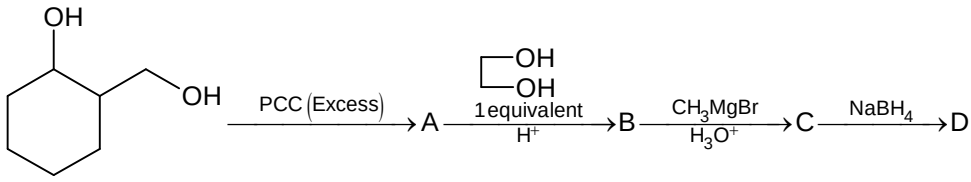
Space for rough work

Chemistry PART – II

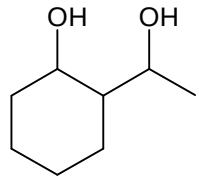
SECTION – A

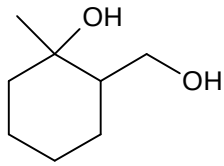
Straight Objective Type

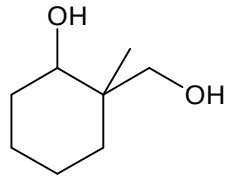
This section contains **6 multiple choice questions** numbered 1 to 6. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.

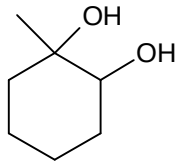
1. 

The product D is:

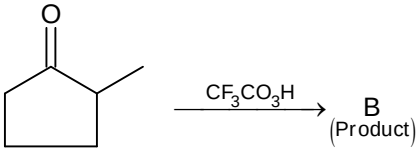
(A) 

(B) 

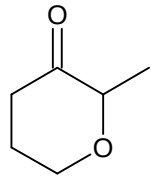
(C) 

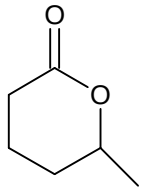
(D) 

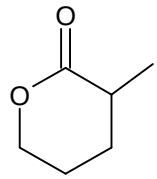
2. In the following reaction,

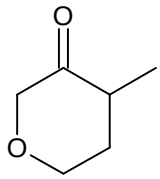


The product (B) is:

(A) 

(B) 

(C) 

(D) 

Space for rough work

3. The standard heat of combustion of $C_2H_2(g)$, C(graphite) and $H_2(g)$ are 310 Kcal, 94 Kcal and 98 Kcal respectively. The heat of formation of $C_2H_2(g)$ is:
(A) 24 Kcal (B) 148 Kcal
(C) 26 Kcal (D) 284 Kcal

4. 20 ml of $\frac{M}{10}$ NH_4OH solution is mixed with 16 ml of $\frac{M}{10}$ HCl solution. The pH of the resulting solution is:(pK_b of NH_4OH = 4.74)
(A) 9.26 (B) 6.85
(C) 4.74 (D) 8.65

5. Which of the following statement is incorrect?
(A) $HClO_4$ is a stronger acid than HClO (B) HF is a stronger acid than HI
(C) BF_3 is a weaker Lewis acid than BCl_3 (D) NF_3 is a weaker Lewis base than NCl_3

6. Which of the following statement is incorrect?
(A) $LiHCO_3$ does not exist in the solid state.
(B) Potassium superoxide is paramagnetic in nature.
(C) Solubility of $Ba(OH)_2$ in water is more than the solubility of $Mg(OH)_2$.
(D) Li_2CO_3 is more stable than Rb_2CO_3.

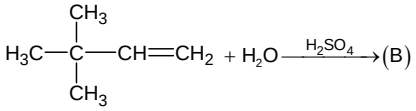
Comprehension Type

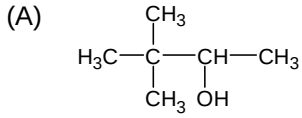
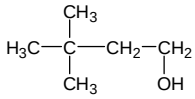
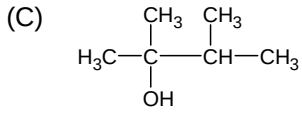
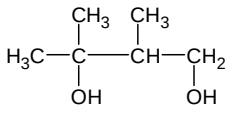
This section contains **2 groups of questions.** Each group has 3 multiple choice question based on a paragraphs. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** is correct.

Paragraph for Question Nos.7 to 9

Acid catalysed hydration of alkene gives alcohol. The addition is in accordance with Markownikov's rule. Alcohols can also be prepared by the oxymercuration-demercuration of alkenes. When alkenes undergo hydroboration oxidation alcohol is formed. In oxymercuration-demercuration and hydroboration oxidation, the net result is the addition of H_2O to the double bond producing alcohol.

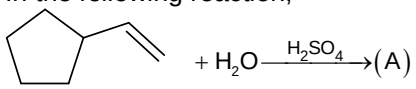
7. Which of the following is the major product (B) in the following reaction?



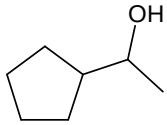
(A) 
(B) 
(C) 
(D) 

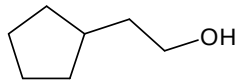
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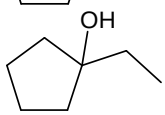
8. In the following reaction,

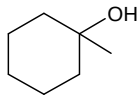


The major product (A) is:

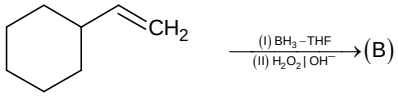
(A) 

(B) 

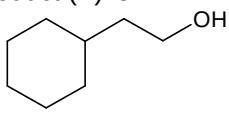
(C) 

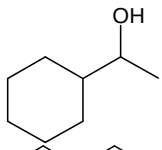
(D) 

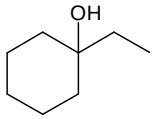
9. In the following reaction:

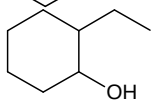


The product (B) is:

(A) 

(B) 

(C) 

(D) 

Paragraph for Question Nos. 10 to 12

A solution is 0.1 M in each KCl, KBr and K_2CrO_4. To this solution solid $AgNO_3$ is gradually added. Assume no change in the volume of the solution.

Given $K_{sp}(AgCl) = 1.7 \times 10^{-10}$

$K_{sp}(AgBr) = 5 \times 10^{-13}$

and $K_{sp}(Ag_2CrO_4) = 1.9 \times 10^{-12}$

10. The sequence of ions precipitated follows the order?

(A) Cl^-, Br^-, CrO_4^{2-} (B) Br^-, Cl^-, CrO_4^{2-}

(C) Cl^-, CrO_4^{2-}, Br^- (D) CrO_4^{2-}, Cl^-, Br^-

11. The concentration of first ion when the second ion starts precipitating is about:

(A) 0.1 M (B) 0.03 M

(C) 0.003 M (D) 0.0003 M

12. The concentration of $[Ag^+]$ required to start the precipitation of AgBr is:

(A) 5×10^{-12} M (B) 1.7×10^{-9} M

(C) 6×10^{-11} M (D) 4×10^{-8} M

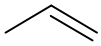
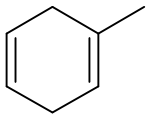
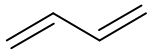
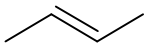
Space for rough work

SECTION – B

(Matching List Type)

This section contains **2 multiple choice questions.** Each question has matching Column(s). The codes for the Column(s) have choices (A), (B), (C) and (D) out of which only **ONE OR MORE is/are correct.**

1. Match the Column – I with Column – II:

Column – I(compounds)	**Column – II(Product(s) of reductive ozonolysis)**
(A) 	(p) OHC – CHO
(B) 	(q) CH_3 – CHO
(C) 	(r) $CH_3-\overset{\overset{O}{\|}}{C}-CH_2-CHO$
(D) 	(s) HCHO
	(t) OHC – CH_2 – CHO

2. Match the Column – I with Column – II:

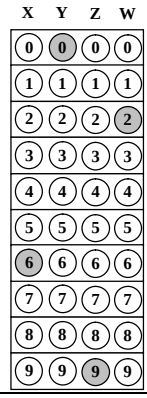
Column – I	**Column – II**
(A) KCl	(p) Antifluorite structure
(B) CsCl	(q) 4 : 4 coordination
(C) ZnS	(r) 6 : 6 coordination
(D) Na_2O	(s) 8 : 8 coordination
	(t) Rocksalt structure

Space for rough work

SECTION – C

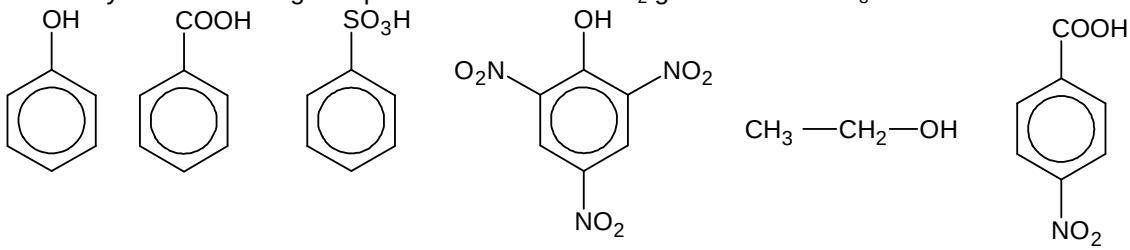
Integer Answer Type

This section contains **5 questions.** The answer to each of the questions is a single-digit integer, ranging from 0 to 9. The appropriate bubbles below the respective question numbers in the ORS have to be darkened. For example, if the correct answers to question numbers X, Y, Z and W (say) are 6, 0, 9 and 2, respectively, then the correct darkening of bubbles will look like the as shown.

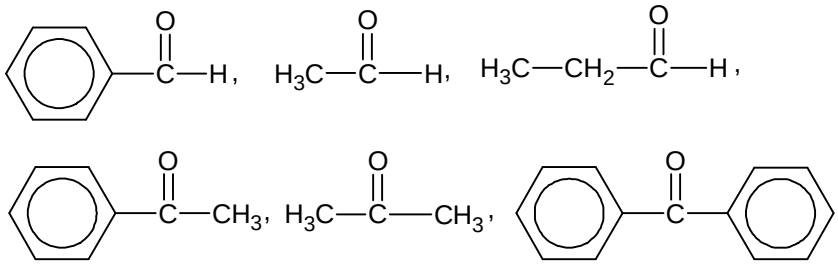


1. Total number of monochlorinated products of $CH_3 - CH_2 - CH_2 - CH_3$ is

2. How many of the following compounds will liberate CO_2 gas from $NaHCO_3$ solution?



3. Among the following the number of compounds which react with Fehling solution is:



4. The number of geometrical isomers of the complex $[Pt(Py)(NH_3)BrCl]$ is?

5. 5 ml of a gaseous hydrocarbon (C_xH_y) was exploded with 30 ml of oxygen. The resulting gas on cooling to room temperature is found to measure 25 ml of which 10 ml are absorbed by NaOH and the remainder by pyrogallol. The value of x + y is?

Space for rough work

Mathematics

PART – III

SECTION – A

Straight Objective Type

This section contains **6 multiple choice questions** numbered 1 to 6. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.

1. The differential equation $\left\{xy^3(1+\cos x)-y\right\}dx+xdy=0$ represents the curve $\frac{x^2}{ay^2}=\frac{x^3}{b}+x^2\sin x+cx\cos x-d\sin x+k$. Then $a+b+c+d$ is equal to

(A) 8 (B) 9
(C) 10 (D) 11

2. Let $f(x)=\int\frac{x-1}{x+1}\frac{dx}{\sqrt{x^3+x^2+x}}$ such that $f(1)=\frac{2\pi}{3}$. Then $f'(1)$ is equal to

(A) 0 (B) $\frac{\pi}{3}$
(C) $\frac{\pi}{4}$ (D) 2

3. If in the triangle ABC, $\frac{AB}{AC}=\frac{3}{4}$ and D is a point lying on BC such that AB $\perp$ AD and D divides BC externally in the ratio 3 : 2, then

(A) $\angle A = 30°$ (B) $\angle A = 75°$
(C) $\angle A = 45°$ (D) $\angle A = 60°$

4. There were two women participating in a chess tournament. Every participant played two games with every other participant. The number of games that the men played among themselves exceeds by 66 that of the games which the man played with the two women then total number of participants is

(A) 11 (B) 13
(C) 10 (D) 12

Space for rough work

5. If the equation $z^3+(3+i)z^2-3z-(m+i)=0$, where $m \in R$ has at least one real root, then m can have the value equal to
(A) 1 (B) 2
(C) 3 (D) 4

6. Find the value of $\sqrt{1+\frac{1}{1^2}+\frac{1}{2^2}}+\sqrt{1+\frac{1}{2^2}+\frac{1}{3^2}}+\sqrt{1+\frac{1}{3^2}+\frac{1}{4^2}}+.....+\sqrt{1+\frac{1}{2010^2}+\frac{1}{2011^2}}$
(A) $2011-\frac{1}{2011}$ (B) $2011+\frac{1}{2011}$
(C) $2011-\frac{1}{2010}$ (D) none of these

Comprehension Type

This section contains **2 groups of questions.** Each group has 3 multiple choice question based on a paragraphs. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** is correct.

Paragraph for Question Nos. 7 to 9

Read the following write up carefully and answer the following questions:

Let $h(x)=\sin^{-1}\frac{1-x^2}{1+x^2}$ and $g(x)=x^4-2x^3+x^2+4$.

$h(x)$ increases in $(-\infty, a_1)$

$h(x)$ decreases in (a_1, ∞)

$g(x)$ decreases in $(-\infty, a_1)$

$g(x)$ increases in (a_1, a_2)

$g(x)$ decreases in (a_2, a_3)

$g(x)$ increases in (a_3, ∞)

$l(AB)$ = portion of tangent at $x = a_1$ to the curve $g(x)$ intercepted between $x = a_1$ and $x = a_3$.

$l(CD)$ = portion of tangent at $x = a_2$ to the curve $g(x)$ intercepted between $x = a_1$ and $x = a_3$.

Join BC and AD with straight line

7. Let number of points of non differentiability of $y=|h(x)|$ be a_4. Then $a_1+a_2+a_3+a_4$ is equal to
(A) $\frac{9}{2}$ (B) $\frac{7}{2}$
(C) $\frac{11}{2}$ (D) 5

Space for rough work

8. Area of quadrilateral ABCD is

(A) $\frac{1}{16}$ (B) $\frac{1}{4}$

(C) $\frac{9}{16}$ (D) none of these

9. Area enclosed by $y = h(x)$ and x-axis is

(A) $1+2\ln 2$ (B) $\frac{\pi}{2}-\ln 2$

(C) $\frac{\pi}{4}+\ln 2$ (D) none of these

Paragraph for Question Nos. 10 to 12

Read the following write up carefully and answer the following questions:

Consider the equation $x^4+2ax^3+x^2+2ax+1=0$, where $a \in R$. Also, range of $f(x)=x+\frac{1}{x}$ is $(-\infty, -2]\cup[2, \infty)$

10. If equation has at least two distinct positive real roots, then all possible values of 'a' are

(A) $\left(-\infty, -\frac{1}{4}\right)$ (B) $\left(\frac{5}{4}, \infty\right)$

(C) $\left(-\infty, -\frac{3}{4}\right)$ (D) none of these

11. If equation has at least two distinct negative real roots, then all possible values of 'a' are

(A) $\left(\frac{3}{4}, \infty\right)$ (B) $\left(-\frac{5}{4}, \infty\right)$

(C) $\left(-\infty, \frac{1}{4}\right)$ (D) none of these

12. If exactly two roots are positive and two roots are negative, then number of integral values of 'a' is

(A) 2 (B) 1

(C) 0 (D) 3

Space for rough work

SECTION – B

(Matching List Type)

This section contains **2 multiple choice questions.** Each question has matching Column(s). The codes for the Column(s) have choices (A), (B), (C) and (D) out of which only **ONE OR MORE is/are correct.**

1. If a, b, c are sides of a triangle and a^3, b^3, c^3 are roots of $x^3 - px^2 + qx - r = 0$. Then, match the following Column–I with Column–II

Column – I		Column – II	
(A)	$\sin^3 A + \sin^3 B + \sin^3 C - 3\sin A \sin B \sin C = 8\Delta^3$	(p)	p = 14, q = 5, r = 8
(B)	$a\sin^2 A + b\sin^2 B + c\sin^2 C = 14\Delta^2$	(q)	p = 36, q = 7, r = 27
(C)	$\sin A \sin B \sin C = 2\Delta^3$	(r)	p = 9, q = 7, r = 8
(D)	$a\cos 2A + b\cos 2B + c\cos 2C = 2\left(s - \Delta^2\right)$	(s)	$p = \frac{63}{2}$, q = 5, r = 27
		(t)	p = 1, q = 8, r = 8

2. Let $f(n, k)$ be the probability that when n coins are tossed, then k heads are obtained $(k \le n)$. Match the following Column–I with Column–II

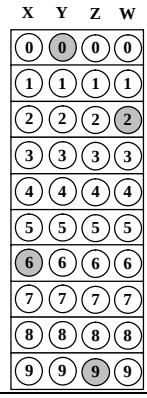
Column – I		Column – II	
(A)	$f(n, k)$ when no 2 heads are consecutive $(n \ge 2k-1)$	(p)	$\frac{2^n - {}^{n+k+1}C_k}{2^n}$
(B)	$f(n, k)$ when k heads occur together	(q)	$\frac{2^n - {}^{n-k+1}P_1}{2^n}$
(C)	$f(n, k)$ when all k heads do not occur together	(r)	$\frac{{}^{n-k+1}C_k}{2^n}$
(D)	$f(n, k)$ when 2 heads are consecutive	(s)	$\frac{{}^{n-k+1}P_1}{2^n}$
		(t)	$1 - \frac{{}^{n-k+1}C_1}{2^n}$

Space for rough work

SECTION – C

Integer Answer Type

This section contains **5 questions.** The answer to each of the questions is a single-digit integer, ranging from 0 to 9. The appropriate bubbles below the respective question numbers in the ORS have to be darkened. For example, if the correct answers to question numbers X, Y, Z and W (say) are 6, 0, 9 and 2, respectively, then the correct darkening of bubbles will look like the as shown.



1. A semicircle is drawn outwardly on chord AB of the circle with centre O and unit radius. The perpendicular from O to AB meets the semicircle on AB at C. Then length of AB so that OC has maximum length is $\sqrt{K}$. Then K is equal to _____

2. In a ΔABC, $\angle B = 60^o$. Then, one of the roots of the equation $(a-c)^2 x^2 - b^2 x + ac = 0$, where a, b, c are the sides of the triangle, is equal to _____

3. The centre of a circle passing through foci of a hyperbola $\frac{x^2}{9} - \frac{y^2}{4} = 1$ and the points at which the tangent at the point (6, $2\sqrt{3}$) meets the tangents at the vertices is $\left(h, \frac{-k}{\sqrt{3}}\right)$, then k is equal to _____

4. There are 15 seats in the first row of a theatre of which 6 are to be occupied. The number of ways of arranging 15 person so that no two persons are side by side is k + 5, then digit in unit's place in k is equal to _____

5. The remainder when $6^{83} + 8^{83}$ is divided by 49 is equal to _____

Space for rough work